M13-C3 — H4b cautious split under leading a target c on the H3 fixture

H4b run / analysed / not a Bucket 3 claim

1 Status

H4b run / analysed. Bounded follow-up to H3/H4: repository `baseline_cautious` vs locked H3 ECAI brave on `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, focus target c. Markdown twin: `h4b_cautious_split_under_a.md`.

Lead (Layer B): cautious **only** changes the α_3 close — rejects brave residual `c_alpha_3 :- alpha_1, a_val_1` and accepts `c_alpha_3 :- d_val_1`; both arms `solved`. **Shared (Layer A — not a cautious effect):** both arms still gate on BK-leading a and keep the α_1-nest vs α_2-flat contrary split. **Not H5.** H0–H4 closed. **Not a Bucket 3 claim.**

2 Research question

On the H3 BD AND + BK-leading-a table, under otherwise shared nd/`relto` settings, does switching from ECAI brave to repository-baseline cautious change how the split beneath the a gates is closed — in particular the attack on `alpha_3` — and what stays the same?

3 Main finding (Layer B — lead)

Both arms **solve**. The string difference is **exactly one rule** — the body of `c_alpha_3`:

```
% brave:    c_alpha_3(A) :- alpha_1(A), a_val_1(A).
% cautious: c_alpha_3(A) :- d_val_1(A).
```

Decisive cautious fork:

```
folding result: c_alpha_3(A) <- [a_val_1(A)]
 found: alpha_1/1 ... KO, cannot introduce an assumption for [a_val_1(A)]
 folding result: c_alpha_3(A) <- [d_val_1(A)]
```

Locked brave at the same fold: `alpha_1` **OK** → `c_alpha_3 :- alpha_1, a_val_1` (never folds d_* here).

Lead: `baseline_cautious` keeps H3's leading-a gates and the persistent α_1-nest vs α_2-flat contrary split; it **only** changes the α_3 close. Both arms `solved`.

Versus H4: H4 blocked a brave residual gadget on **roots** → `completed_no_solution`; H4b blocks *this* residual gadget on **child** c → still `solved`, with explicit `d_val_1`.

4 Shared background (Layer A — not cautious)

Both arms still gate on BK-leading a (`c :- alpha_1, a_val_1` / `c :- alpha_2, a_val_0`); evaluator BD AND is not recovered; body vars $\{a\}$.

Contrary skeletons (identical through α_3 intro):

Gate	Contrary structure (both arms)
α_1 (a=1)	Nested: `c_alpha_1 :- b_val_0` plus `c_alpha_1 :- alpha_3, b_val_1`
α_2 (a=0)	Flat: `c_alpha_2 :- d_val_0` and `c_alpha_2 :- b_val_0`

Procedural reason (search — not cautious): `c_alpha_1` repaired first; under a=1 commit `b_val_0` then mint α_3 on residuals; under a=0, `relto` KO on existing α_3 forces flat d/b cover. Do **not** credit cautious with symmetrising / removing the nest / fixing the asymmetry.

5 Brave vs cautious table

Layer A objects are the same; Layer B differs only in `c_alpha_3` (`alpha_1,a_val_1` vs `d_val_1`). Both outcomes `solved`.

6 Setup / paths

Same H3 fixture/sample (sha256 `48bd0da0...db05`). Intervention: locked ECAI brave → `baseline_cautious` (not H5). Incidental: collection targets $a/b/d$ timed out at 300s — outside focus.

7 Semantic boundary

Authoritative: Prolog learner outcome under selected `learning_mode`. Under cautious learning, SAT of `bk.sol_chk.asp` is an existential integrity witness only.

8 Boundaries / non-claims

Not a Bucket 3 claim; not H5; not mechanism recovery. Layer A is shared with H3 and is not a cautious effect. H5/H6/H7 unchanged.

9 Next

H4b documentation is complete. Orchestrator / Samuel decides the next move. Still **no Bucket 3 claim**.